Formalizing the IUT III 3.11–3.12 Corridor in Lean

IUT Lean formalization project

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Abstract

This paper reports the current state of a Lean formalization project for a narrow part of Mochizuki's inter-universal Teichmuller theory: the corridor from IUT III, Theorem 3.11 to Corollary 3.12. The project is deliberately neutral about the correctness of IUT as a proof of the *abc* conjecture and about the Mochizuki–Scholze–Stix dispute. Its purpose is to make the comparison levels, indeterminacy actions, real-line transports, and source obligations explicit enough that the proof assistant can distinguish proved consequences from mathematical hypotheses still to be constructed.

The formalization now verifies the ordered-real endpoint of Corollary 3.12, separates the representative j^2-level from full-label, averaged, and hull/log-volume comparison levels, and builds a typed route through concrete Hodge-theater/log-theta-style source data, $\mathrm{Ind}_1/\mathrm{Ind}_2/\mathrm{Ind}_3$indeterminacy records, quotient possible images, a selected q-region, Remark 3.9.5-style hull/determinant data, and the existing C_Θ endpoint. The possible-image layer is now witness-bearing: class-level possible images, quotient pullbacks, and the selected q-region carry explicit nonemptiness data through the Step (xi) valuation-ball/log-shell source chain. The class/log-theta-lattice formula and typed fiber-transport sources now carry these witnesses as part of the same Theorem 3.11 spine; the top-level additive-Haar arithmetic-divisor audit consumes the corresponding nonemptiness proof to build the witness-bearing Theorem 3.11 possible-image record, rather than using a vacuous possible-image predicate. The arithmetic-divisor and record Ob3/Ob5 additive-Haar constructors can now obtain all-choice possible-image nonemptiness directly from the class-indexed Theorem 3.11 image law and the record/source image equality; the same witness bridge constructs the central Theorem 3.11 possible-image source-data record and its nonemptiness audit. The remaining lower task is to construct those source fields from the analytic multiradial/log-theta objects of the papers. The current result is a checked conditional Stage 1 corridor, not a closed proof of Corollary 3.12 from Mochizuki's published hypotheses.

We also maintain a Lean blueprint and intend to make the project consumable by Apodeixis as a collaboration and review layer. The blueprint is presently a compact navigational document plus a generated declaration index; it is useful for orientation, but it is not yet a complete mathematical dependency graph. This paper replaces an earlier ledger-style draft. Future updates should revise the relevant sections of this research paper rather than append chronological implementation notes.

1 Introduction

IUT is large enough that a useful formalization cannot begin by translating all four papers in order. The present project instead focuses on the public pressure point around IUT III, Theorem 3.11 and Corollary 3.12. Mochizuki's 2026 formalization report identifies this as the first stage of a possible formalization program:

$$\text{Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{Corollary 3.12.}$$

The label "3.11.5" is not a theorem in the IUT III paper. It names the reorganized checkpoint obtained by moving the holomorphic hull plus determinant operation before the final simultaneous comparison of the Θ-pilot and the q-pilot.

The immediate formal target is not the *abc* conjecture. It is the logical shape of the Stage 1 implication. The question is:

> If the source constructions claimed by IUT III supply the multiradial possible images, indeterminacy behavior, IPL, SHE, APT, hull/determinant operations, and local-to-global C_Θ estimates, does the Corollary 3.12 inequality follow in a type-correct way?

The current code answers a substantial conditional version of this question. It proves the final ordered-real implication, constructs many intermediate typed routes, and exposes the remaining source mathematics as named obligations. It does not yet prove that the full Hodge-theater, Frobenioid, log-Kummer, holomorphic hull, determinant, and IUT IV analytic estimates of the source papers instantiate those obligations.

2 Mathematical Target

2.1 The final inequality

The endpoint formalized in Lean is the signed real comparison used in the Corollary 3.12 corridor. In the notation of the project, the route must supply

$$q_{\text{signed}} \leq \Theta_{\text{signed}}, \qquad 0 < -q_{\text{signed}}, \qquad \Theta_{\text{signed}} \leq C_\Theta(-q_{\text{signed}}).$$

From these hypotheses the ordered-real calculation gives

$$-1 \leq C_\Theta.$$

The Lean theorem behind this step is elementary. This is important: the hard question is not whether the final real inequality follows from the displayed signed comparisons. The hard question is whether the source machinery really constructs the signed comparison without collapsing distinct histories or comparison levels.

2.2 The disputed comparison level

Scholze and Stix criticize the proof around Corollary 3.12 on the grounds that the relevant Θ-side data carries j^2-scaled information, while a naive simplification of the comparison appears to remove the room needed for that data. The formalization responds by enforcing separate types for:

representative j^2-coordinate data,
balanced sign-compatible data,
full-label and averaged finite-label data,
transported ordered-real copies,
hull/log-volume endpoint data.

Lean then rejects the most naive collapse: the balanced sign-compatible layer does not by itself supply the final Corollary 3.12 route. This is a diagnostic result, not a settlement of the dispute. The remaining issue is whether the source construction gives the final hull/log-volume object with the required compatibilities.

2.3 Indeterminacy behavior

The current Stage 1 interface treats the three indeterminacy components separately. In the formal model, Ind_1 and Ind_2 preserve procession-normalized log-volume and generate equality for the possible-image quotient. Ind_3 is not an equality generator; it contributes only an upper-semi log-volume inequality. This distinction is now part of the checked Lean interface:

$$\mathrm{Ind}_1, \mathrm{Ind}_2 : \text{equality/invariance}, \qquad \mathrm{Ind}_3 : \text{upper-semi inequality}.$$

This separation is one of the central design decisions of the formalization. It prevents a proof from turning an upper-semi source relation into an equality merely because both appear in the informal phrase "indeterminacy".

3 Formalization Architecture

3.1 Source, Lean, paper

The working cycle is:

$$\text{Mochizuki source papers} \longrightarrow \text{Lean definitions and theorems} \longrightarrow \text{research paper and blueprint.}$$

The papers in `docs/` are treated as source of truth. The Lean code is the formal artifact. This paper records the mathematical state of the formal artifact; it is not meant to be a chronological log. This distinction matters because previous versions of this draft had grown into an implementation ledger. The present version is organized as a research paper: target, architecture, results, trust boundary, tooling, and next work.

3.2 Main Lean layers

The Stage 1 development is split into modules whose roles are stable enough to describe at paper level:

Module family	Role
`PilotComparison`, `CorollarySchema`, `SourceObligations`	The signed real endpoint, the final Corollary 3.12 predicate shape, and the ledger turning a common-bound comparison into the endpoint.
`IUTStage1SourceCore`, `IUTStage1FiniteLabels`, `IUTStage1Gaussian`	Real-line copies, regions, finite-label and $ZMod(\ell)$ models, Gaussian degree shadows, and j^2-diagnostic infrastructure.
`IUTStage1StepX`, `IUTStage1HodgeSHE`	Procession-normalized log-volume routes, Hodge/SHE data, square/full-label transport, and upper-semi compatibility records.
`IUTStage1Theorem311`, `IUTStage1StepXI`	Typed Theorem 3.11 indeterminacy cores, possible-image quotient structures, concrete Hodge-theater/log-theta source layers, Remark 3.9.5 hull/determinant interfaces, and paper-trace audits.
`IUTStage1FrobenioidShift`, `IUTStage1Experiments`	Nonarchimedean realified Frobenioid/log-Kummer and vertical-IQ routes, additive-Haar arithmetic-degree endpoints, and compiled public diagnostics.

The names in this table are intentionally coarse. A paper should not reproduce thousands of declaration names. The detailed declaration surface is maintained in the blueprint's generated declaration index.

3.3 Records as boundaries

The project uses records in two different ways. Some records are ordinary mathematical data structures whose fields are proved or constructed in Lean. Other records are trust-boundary interfaces: they name a source construction that must eventually be derived from the IUT papers. The current paper distinguishes these roles explicitly.

For example, the typed indeterminacy core is a useful formal object: it stores the $\mathrm{Ind}_1/\mathrm{Ind}_2$ invariance laws, the Ind_3 upper-semi law, and the quotient relation. By contrast, the full Theorem 3.11-and-remarks obligation record still contains source-facing fields such as construction of the input prime-strip link, theta-pilot possible images, the log-theta-lattice procession, and the Hodge/SHE/IPL/APT bridge. These fields are not hidden axioms, but they are also not yet source-derived mathematics.

4 Current Formal Results

4.1 Ordered-real endpoint

The final ordered-real implication is formalized. Once the signed comparison $q_{\text{signed}} \leq \Theta_{\text{signed}}$, positivity of $-q_{\text{signed}}$, and the C_Θ-upper bound are available, Lean derives the expected lower bound $-1 \leq C_\Theta$. The code also proves the corresponding strict/boundary dichotomy: either the boundary case is exact, or $-1 < C_\Theta$.

This part is no longer an open mathematical problem inside the project. It is pure ordered-real algebra. All serious mathematical work sits upstream of the signed comparison and the C_Θ-upper bound.

4.2 Anti-collapse diagnostics

The formalization proves negative diagnostic statements about comparison-level collapse. In particular, a label-independent j^2 collapse is rejected at the representative level, and the balanced sign-compatible layer is not accepted as the final route level. These diagnostics are valuable because they test the exact failure mode emphasized in external criticism.

The diagnostics should be read carefully. They show that the Lean development does not silently make the criticized simplification. They do not show that Mochizuki's construction successfully supplies the stronger hull/log-volume data.

4.3 Concrete Stage 1 route

The current strongest route starts with a unified Hodge-theater/log-theta/log- Kummer source spine. This spine carries the 0- and 1-column histories, prime-strip data, theta-pilot possible images, the selected q-choice, and the vertical log-Kummer packet alignment. Its Theorem 3.11 bridge side is also structured: the input-prime-strip link, one-column simultaneous holomorphic expression, and algorithmic parallel transport are projections of one IPL/SHE/APT bridge package. The Step (x) side is represented by a single finite-divisor/vertical-log-Kummer package: finite divisor tensor packets, realified Frobenioid Kummer compatibility, mono-analytic forgetting, vertical IQ, and source-target log-volume calibration are projections of one object rather than five unrelated source flags. Lean projects from the same spine both the Theorem 3.11 typed $\mathrm{Ind}_1/\mathrm{Ind}_2/\mathrm{Ind}_3$ core and the Step (x) payload. The route then forms the equality-quotient family of possible images, selects the q-region, and connects that region to the existing C_Θ/Corollary 3.12 corridor. The preferred paper-trace route now has a named source-spine audit: the Theorem 3.11/Remarks obligation package and the Step (x) finite-divisor/log-Kummer obligation package are projected from one source object. The first Step (xi) hard-math route has also been added: a paper-derived Remark 3.9.5/Step (xi) source records the substeps (xi-a)–(xi-g), ties the selected q-region to the actual Theorem 3.11 possible image selected by the source spine, and projects the existing Step (xi) hull/determinant obligation record internally. Ob7 is no longer only a raw compatibility flag on the newest route: the legacy prime-strip/log-Kummer record and its input/output strip and selected-column equalities are projected from a coric $\Theta^{\times\mu}$/log-Kummer source link aligned with the same quotient hull source. The preferred localized route now exposes this source as a structured audit: q-region containment, Gaussian/environment place agreement, coric unit-character agreement, and the three hull-bridge alignments are checked before the legacy compatibility record is projected. The constructed-paper-trace public boundary now also packages this information together with the constructed Step (xi-a)–(xi-g) and Ob1–Ob9 ledger, making explicit that Ob7–Ob9 are read from the same log-Kummer/bridge source rather than supplied as separate trace facts. Its strictest boundary now derives the non-Step-(xi) audit from the proof-carrying Theorem 3.11, Step (x), and IUT IV source records; the caller supplies only the explicit additive-Haar arithmetic-degree payload that remains below this milestone. The selected q-region containment is now derived from the equality-quotient possible-image family: it is the canonical inclusion of the selected possible region into the family union, followed by the constructed holomorphic hull source. The possible-image family is no longer allowed to be an empty predicate shell at this boundary: the theta-pilot class image source carries a point witness for every class, Lean pulls these witnesses back to concrete choices, and the equality-quotient construction proves nonemptiness of the quotient pullbacks and of the selected q-region. The Step (xi) valuation-ball/log-shell source chain now transports the same theta-region witnesses down to the local exact-source constructors. More recently, the class-region, class-formula, log-theta-lattice formula, lattice-image-law, and typed fiber-transport source records themselves were made witness-bearing, so this nonemptiness data is no longer introduced independently by the valuation wrapper. The paper-derived Step (xi) source also has a quotient-compatible constructor that fixes the selected choice to the source spine, instead of accepting an independently assembled hull record at that point. The remaining IUT IV and additive-Haar layers are then combined with this constructed Step (xi) source. A goal-facing completion audit packages this milestone into checked claims:

paper trace, concrete choice space, typed indeterminacy core,
procession/F_ℓ, quotient possible images, selected q-region,
Remark 3.9.5 bridge, constructor into the corridor,
lowered endpoint, audit/paper handoff.

The audit is a projection of the assembled remaining-payload record. It does not add a new mathematical assumption. Its function is to make the project milestone checkable and reviewable.

4.4 Local-to-global C_Θ chain

The code now contains a proof-carrying local-to-global C_Θ route at the interface level. The preferred additive-Haar arithmetic-degree and p-adic formula route threads local analytic estimates, arithmetic-degree calibration, finite-place Haar defects, formula-gap matching, and Step (xi) local terms into the endpoint. For this preferred route, the canonical C_Θ-scale comparison is derived from the finite-place arithmetic gap and the IUT IV handoff identity, rather than supplied as a raw endpoint hypothesis. At the Step (xi) source layer, Lean now also proves the sharper handoff for both the constructed Remark 3.9.5 hull/determinant source and the constructor-built exact-theta source: once the local canonical Step (xi) scale is identified with the finite-place scale in the IUT IV summation record, the global C_Θ-dominance follows from the IUT IV plus-one cancellation theorem.

The phrase "proof-carrying" is deliberate. The route no longer consumes a single unstructured numerical inequality at the public endpoint. It consumes a structured chain of local and global source objects. Nevertheless, several of those source objects still stand for arithmetic constructions not yet derived from number-field valuations, conductors, differents, holomorphic hulls, and the local estimates of IUT IV.

5 What Remains Mathematically Open

5.1 Theorem 3.11 from source data

The largest remaining mathematical task is to construct the Theorem 3.11 multiradial output from actual IUT I–III source data. The present code has a unified Hodge-theater/log-theta/log-Kummer source spine that projects to the typed core, possible-image quotient, input-prime-strip link, SHE/IPL/APT bridge, and Step (x) finite-divisor/log-Kummer payload. The preferred paper-trace API no longer asks for those Theorem 3.11 and Step (x) pieces as parallel flags; they are read from the spine. The implementation does not yet derive that spine, or the structured bridge and Step (x) packages inside it, from the full Hodge-theater, local LGP, and Frobenioid constructions. The missing source construction includes:

initial theta data and Hodge-theater histories,
prime strips and the derivation of the structured input link,
the multiradial algorithm and its possible images,
the log-theta-lattice procession and F_ℓ-symmetry,
the derivation of the 1-column SHE/IPL/APT bridge,
the derivation of the Step (x) package from vertical log-Kummer data,
the proof that $\mathrm{Ind}_1/\mathrm{Ind}_2$ preserve normalized log-volume,
the proof that Ind_3 has only the stated upper-semi orientation.

The Lean records now say exactly what must be supplied. They do not yet replace Mochizuki's proof of Theorem 3.11.

5.2 Remark 3.9.5 and Step (xi)

The second major gap is the genuine holomorphic hull and determinant construction. The current code now has a paper-traced Step (xi) source route: the old Step (xi) obligation record can be projected from a structured source that records Corollary 3.12 Step (xi-a)–(xi-g), Remark 3.9.5 Ob1–Ob9, the selected Theorem 3.11 possible image, determinant data, and Ob7 link pinning. At the hull-formation boundary, the selected q-region is now constructed from the equality-quotient possible-image family, so its containment in the possible-image union is no longer a separately supplied Step (xi) field; the paper-derived source constructor uses the source spine's selected q-choice directly. This boundary has now been lowered one step further: the Step (xi) hull source may be built from the Remark 3.9.5 minimal-hull system $\phi(P) = \bigcap_{P\subseteq H} H$. Lean derives the holomorphic hull operator, the canonical hull of the possible-image union, closedness of this canonical hull, containment of the selected q-region in the union, and the Ob1/Ob2 absorption statement from this minimal-hull source rather than from a free hull-operator witness. The current lower constructor also accepts the principal product presentation $\lambda \cdot O$ of Remark 3.9.5(i): the base integer/valuation-ball region O, nonzero scalar parameters, scalar-image principal hulls, the intersection-parameter hull, and the principal-hull log-volume are projected into the concrete Step (xi) hull source. At the determinant boundary, the Step (xi) determinant comparison record can now be constructed from the weighted arithmetic-vector-bundle determinant source already present in the Step (xi) layer: the finite adjusted summands, determinant sum, normalization, and positive tensor power are projected from that lower source rather than supplied as independent fields. This has now been lowered to the localized arithmetic-vector-bundle decomposition source: the localized hull cover supplies a finite sum of localized hull-region log-volumes, each localized log-volume is identified with the corresponding structure-sheaf-adjusted vector-bundle contribution, and Lean projects the result to the Step (xi) determinant comparison data. The selected q-region-to-normalized-determinant inequality is then derived from hull containment plus the localized family-hull/normalized-determinant equality; the localized projection audit now also exposes the adjusted Ob3/Ob5 endpoint, the induced Ob3/Ob5 compatibility source, and the bounded-family hull/determinant log-volume endpoint proved in the lower SourceCore layer. The strict localized route now packages this with the equality-quotient Ob5 statement: choices in the $(\mathrm{Ind1}, \mathrm{Ind2})$ equality orbit have identical quotient possible regions and quotient-region log-volumes, while the same localized family-hull construction supplies the determinant and bounded-family tensor endpoint. The record Ob3/Ob5 boundary has also been lowered through the theta-region-defined principal valuation-ball source to the typed fiber-transport valuation source. Lean pulls the theta-pilot valuation cover back to concrete choices, proves that its adjusted possible-region family is the multiradial record possible-image family, builds the record Ob3/Ob5 adjusted determinant/log-volume source from this cover, and then feeds the resulting source into the exact-theta constructed hull/determinant API. For a lattice-image-law-backed source the remaining explicit input is the identification of the supplied lattice-image law with the concrete fiber-transport law; for the fiber-transport-backed source this equality is internal to the constructor. This boundary has also been lowered through the local log-shell/log-Kummer layer: a principal pointwise constructed log-shell metric-zero source now constructs the same record Ob3/Ob5 source, and compact-open log-Kummer image, correspondence, and map possible-region sources construct both the record Ob3/Ob5 source and the exact-theta Step (xi) hull/determinant source. Lean passes through the local log-Kummer graph or relation-level image laws, compact-open representatives, and the converse realization law before entering the principal valuation-ball constructor. The correspondence route now also has an obligations-backed entry point: the package bridge equality, q-pilot positivity, and normalization are projected from 'IUTStage1SourceHullDetObligations'. Its Hodge-calibrated variant removes the raw theta/family-hull scalar equality from this boundary: Lean derives Θ_{signed} = family-hull log-volume from 'IUTStage1HodgeFamilyHullLogVolumeCalibration', via the Hodge–Arakelov theta monoid degree. This Hodge-calibrated obligations route now also has a compact-open log-Kummer map entry point: the realization map and local sections construct the correspondence relation internally before the Hodge-calibrated Step (xi) constructor is invoked. The record-canonical bridge-data synchronization and measure calibration remain explicit lower inputs at this boundary. The same calibration now reaches the finite-divisor constructor-backed source: Lean converts the Hodge-calibrated data to the constructor-backed spine, forms the constructor-generated hull/determinant obligations, and builds the possible-image Step (xi) source without reintroducing a raw Θ_{signed}-to-family-hull equality at this boundary. The target-charted exact-theta adjusted-summand source now enters this Hodge-calibrated finite-divisor constructor first and projects the older constructor-backed source only for legacy consumers. The all-in-one target-charted Hodge/IPL possible-image route audit also exposes the record-canonical Theorem

3.11 hull/tensor-power bridge equality beside the ambient-obligations bridge equality, making this synchronization visible at the public route boundary instead of only through the obligations projection. It now also carries the target-charted reduction to the constructor-built Step (xi) source and the resulting Remark 3.9.5 Ob1–Ob5 bridge audit, so the public route records the actual possible-image hull absorption and determinant-compatibility chain below the canonical bridge equality. The same route now has a parameterized Ob5 quotient audit: for any nonempty comparison possible image, it projects the constructor-built bounded-family hull inclusions, collapsed upper-semi quotient equality, determinant bound, and local canonical-scale chain. The same public route now exposes the Ob6 exact-$\Xi(P_B)$ boundary. It constructs the Remark 3.9.5 possible-image family from the constructor-built hull source, identifies the canonical $\Phi(P_B)$ approximant with the family hull, and derives the selected and comparison log-volume bounds from a source-calibrated exact-$\Xi(P_B)$ log-volume equality rather than from an arbitrary supplied approximant family. This target-charted route boundary now also has a constructor-built Ob7 product-formula source. The input is finite weighted determinant prime-strip/product-formula data plus the coric unit-character pullback; Lean requires its determinant source to be the same determinant source used by the constructor-built Step (xi) bridge, constructs the coric log-Kummer Ob7 source, and then packages the combined Ob5/Ob6/Ob7 audit. Thus the route no longer needs a standalone determinant/global Ob7 equality at this level, although the lower local product-formula and coric prime-strip data remain explicit source inputs. At the principal-product localized public route, the Step (xi)/Remark 3.9.5 paper trace is now also constructed internally. Its Step (xi-a)–(xi-g) and Ob1–Ob9 entries are concrete statements about the unified Theorem 3.11 source spine, the $\lambda \cdot O$ hull source, the localized determinant projection, the theta-equals-family-hull equality, and the shared Ob7 log-Kummer source. The strict public route therefore no longer accepts a standalone paper-trace proposition bundle. A localized-cover variant now lowers the determinant alignment boundary: its source fixes the hull system and possible-image family to the principal-product Step (xi) source, so Lean projects the localized decomposition with the old hull-operator and possible-region equalities by reflexivity rather than accepting them as separate public inputs. Its canonical-theta refinement also takes Θ_{signed} to be the projected family-hull log-volume, making the theta/family-hull equality reflexive at this boundary. At the Additive-Haar local C_Θ boundary, the constructed Remark 3.9.5 hull/determinant source can now be built directly from the localized vector-bundle determinant source; hence the determinant-source handoff to the Step (xi) local-term constructor is reflexive. The remaining equality at this local boundary is the genuine calibration of the canonical C_Θ scale with the finite Step (xi) sum. The same constructed-trace handoff has now been propagated to the finite-extension valuation-unit-ball compact-open norm-square boundary, using the localized determinant, Hodge-evaluation, theta/family-hull, and Ob7 construction chain already required by that strict route. A calibrated variant of this boundary now replaces the raw Hodge theta-monoid/localized-adjusted sum equality by the Hodge/family-hull equality and derives the adjusted-sum equality from the localized Ob3/Ob5 identity. A further restriction-calibrated variant replaces the placewise Hodge canonical-degree/local-prime equalities by an anchored local-global restriction source together with one global Hodge canonical-degree calibration. The strongest finite-extension variant now also moves the remaining localized possible-region comparison down to the p-adic finite source itself: the equality after the unit-ball/additive-Haar/compact-open/norm-square projections is derived by the source-core projection theorem. The principal-product finite-extension source lowers this once more: its localized possible-region field is fixed definitionally to the Theorem 3.11 principal-product possible-image family, so the strongest public wrapper no longer takes a p-adic possible-region equality as an independent input. A calibrated p-adic finite hull-cover source now also derives the pointwise Ob3 localized-region log-volume identities from local vector-bundle calibrations, instead of accepting those identities as fields of the finite-extension localized decomposition source. Its family-hull-indexed refinement derives the cover identity, pairwise disjointness, and finite-additive volume sum from an index map on the principal-product family hull. A cover-local measure-model refinement lowers the p-adic volume input further: the localized cover sum is derived from the log-volume measure model attached to the calibrated p-adic regions, and the preferred public finite-extension route can consume this measure-backed cover source directly. A direct-product refinement removes the supplied p-adic family-hull index: Lean constructs it from the local-factor cell cover of the principal-product family hull, using local-factor separation to prove the fiber and disjointness properties. A tensor-measure refinement removes the supplied p-adic cover-local measure model: direct-product cell log-volume identities and their finite tensor-cell summation formula construct the measure model used by the preferred public route. The p-adic tensor data has also been lowered to the source-core localized decomposition: Lean derives the tensor-cell log-volumes and tensor-sum formula from the adjusted determinant log-volume identities carried by that finite-extension hull/vector-bundle source. The direct-product local factors in this p-adic branch are now charted by local-ring/vector-bundle factor calibrations, so Lean projects their local-ring equality and direct-summand log-volume identities instead of taking raw factor regions at the public boundary. These chart calibrations have in turn been lowered to valuation-ball factor calibrations: the p-adic local-factor regions are now supplied by valuation-ball/additive-Haar data and then projected to the charted local-ring interface. A further principal-product compact-open-factor variant replaces the direct Hodge/family-hull equality by a localized adjusted-determinant sum calibration; Lean recovers the family-hull equality from the localized Remark 3.9.5 decomposition theorem. The additive-Haar principal-product branch now also has a cover-derived localized hull source: instead of taking the localized-region containment in the family hull and the family-hull log-volume sum as standalone fields, it assumes the explicit cover identity $\phi(\bigcup_i \Theta_i) = \bigcup_\beta H_\beta$ and the finite additivity formula for this cover; Lean derives the old containment and volume-sum interface from these lower hull-cover statements. This boundary has now been lowered again: each H_β carries a local vector-bundle calibration, is presented as a fiber of an index map on the family hull, and Lean derives the cover identity, pairwise disjointness, finite-additive cover sum, and pointwise $\log \mathrm{vol}(H_\beta)$ determinant identity after identifying that local vector bundle with the additive-Haar determinant localization. The strict positive determinant contribution has now been separated from the valuation-unit-ball normalization. A principal-product additive-Haar route constructs the paper trace internally and derives the positive summand from the source-level inequality $1 < \mu(U)$. A stronger local-field branch now constructs such a strict factor explicitly: for a finite extension of $\mathbb{Q}_p$, the inverse base-prime dilation of the normalized unit ball is identified as $p^{-1}O_K$, and the source-core Haar-modulus theorem proves $\mu(p^{-1}O_K) = p^{[K:\mathbb{Q}_p]} > 1$. Conversely, Lean proves that the older unit-ball-positive boundary is contradictory: the same unit-ball source forces $\mu(O_v) = 1$, hence normalized Haar log-volume 0. The principal-product hull source has also been connected to the lower upper-semi quotient formalism of Remark 3.9.5(v): the concrete hull source now projects to a bounded-family hull quotient source, proves that this quotient hull is the canonical/principal family hull, and proves that any two nonempty quotient-indexed possible regions have the same collapsed quotient image. For equality-orbit choices, Lean packages this with region equality, log-volume equality, quotient-image equality, and containment in the family hull. The principal-product source boundary has been lowered again in source-core: a scalar-parameter direct-product hull source now constructs the full Remark 3.9.5 principal product-hull system and its minimal intersection theorem. Product-parameter representability is no longer a global assumption: Lean derives it from coordinate scalar-image data. Once local coordinate points can be extended to product points, the coordinate landing/preimage laws prove that each coordinate product region is the image of the local integer factor under the coordinate scalar map. Lean then uses the coordinate region itself as the local scalar parameter and assembles these choices into the product scalar parameter. The intersection scalar selector is obtained from the represented direct-product intersection parameter. This replaces the principal intersection law by the direct-product hull theorem plus explicit coordinate scalar-image data. This scalar construction has now been transported onto the public Step (xi) carrier: a generic equivalence transport theorem carries principal $\lambda \cdot O$ hull systems across carrier equivalences, and the transported scalar-parameter source constructs the Point(target)-level principal product hull source consumed by the localized theta-equals-family-hull route. Thus the generic route no longer needs a standalone principal product-hull source on the public carrier; it needs coordinate scalar-image data and an explicit carrier equivalence. In the one-coordinate real-line specialization, even this carrier equivalence is not supplied: Point(target) is canonically equivalent to its real coordinate and to the product carrier Unit $\to \mathbb{R}$. For the selected valuation-ball family-hull route, the implementation now uses the more faithful scalar boundary: a nonzero local multiplication source $\lambda_v \cdot O_v$ constructs the selected scalar-parameter valuation source, and the unit-ball valuation wrapper inherits this selected source. The resulting endpoint identifies the selected scalar image with the valuation-ball direct-product cell union and with the adjusted Ob3/Ob5 determinant handoff. The same selected-scalar source now constructs the possible-image exact-$\Xi(P_B)$ source from progressively local cover data: indexed cell containment, single local-factor containment, and finally compact-open valuation-ball-radius containment, where the last step uses the additive-Haar normalization audit $B_{\beta,v} = B_{\beta,v}(r_{\beta,v})$. The nonzero-scalar and valuation-unit-ball wrappers inherit this radius-to-exact-Ξ route. Thus the exact-Ξ calibration is no longer an independent field at this boundary; the remaining public-route obligation is to construct those valuation-ball containments from transported theta/valuation local exactness. For the stricter localized route, the final normalized localized determinant bound by Θ_{signed} is no longer supplied as a standalone inequality: Lean derives it from the identification of Θ_{signed} with the constructed localized family-hull log-volume. The record-canonical Remark 3.9.5 hull/determinant bridge now also feeds this record: once the paper-derived selected q-region log-volume is identified with the record bridge q-pilot log-volume, the q-region-to-normalized- determinant bound and the determinant-to-Θ_{signed} bound are both projected from the bridge. The bridge-backed paper-source constructor now derives this log-volume identification from lower alignment data: equality of the paper hull operator with the bridge hull operator, and equality of the bridge q-pilot region with the source-spine selected Theorem 3.11 possible image. The constructed-source version projects the bridge and hull operator from the constructed Remark 3.9.5 source itself, leaving only the q-pilot region/source-spine selected possible-image identification at this boundary. A further source-selected constructor forms the constructed Remark 3.9.5 source with the source-spine selected possible image as its q-pilot region, so this identification is definitional on that route; the remaining explicit condition is containment of the selected possible image in the record possible-image union. This selected-source boundary now also has a stricter Ob3/Ob5 determinant form: instead of receiving a raw hull/determinant compatibility record, it can receive the record-canonical family-hull compatibility source, or the still lower adjusted determinant log-volume source, and Lean derives the old compatibility projection internally. The pointwise source-region route and the current deepest retained (Ind2) action-packet transport route both have adjusted log-volume variants, so the public local source spine can pass through the measured adjusted-source boundary. That measured boundary now has a Hodge-calibrated constructor: the package measure equality is projected from the measured source, and the Θ_{signed}-to-family-hull equality is derived from ‘IUT-Stage1HodgeFamilyHullLogVolumeCalibration‘, instead of being supplied as a raw downstream argument. This is still a boundary above the localized Ob3/Ob5 arithmetic-vector-bundle data used by the older compatibility routes, but the theta equality at that boundary is now Hodge-derived rather than asserted. The quotient-hull side of these variants has also been lowered: the route may now take the Theorem 3.11 possible-image quotient compatibility for the source images, and Lean forms the Remark 3.9.5 equality-quotient hull/log-volume compatibility by applying the selected holomorphic hull operator. Thus the deepest local selected route no longer needs to expose a standalone equality-quotient hull compatibility record when the lower possible-image quotient source is present. This has now been pushed down to the unified Theorem 3.11 source spine: the route can derive the possible-image quotient compatibility from the theta-pilot possible-image source together with the vertical log-Kummer packet-aligned finite-procession source. On this route the explicit quotient-compatibility argument disappears; the remaining source-side comparison is the record/source possible-image trace supplied by the action-packet transport layer. The determinant upper-bound input has also been lowered on the same source-spine routes: the Ob4 tensor-power bound is derived from the exact-theta calibration identifying Θ_{signed} with the record Ob3/Ob5 family-hull log-volume. Thus the strict action-packet route now uses adjusted determinant log-volume data plus exact family-hull theta calibration, rather than a standalone tensor-power estimate. The Ob7 prime-strip/log-Kummer record can now be read from a single coric $\Theta^{\times\mu}$-style source link aligned with the same quotient-hull bridge; its retained statement includes the local Gaussian/environment equalities and unit-character equalities from the Ob7 source endpoint. The strongest selected route now constructs this source link from the constructed Remark 3.9.5 bridge and a coric prime-strip invariant, after transporting determinant compatibility across the record-union/equality-quotient-family alignment. In the same-index source-spine case, this alignment is now derived from the equality of the record possible-image family with the Theorem 3.11 source image family; the selected route also derives the selected-q containment in the record union from the same equality. A lower selected route now takes the paper-shaped pointwise trace that the record possible image attached to each concrete Hodge-theater/log-theta choice equals the corresponding source-spine possible image; Lean assembles this trace into the full family equality before applying the quotient-union theorem. The selected Ob3/Ob4 route, the adjusted Ob3/Ob5 source-region route, and the exact-theta source-spine route now also have product-formula Ob7 variants: the determinant/global prime-strip equality is proved from finite local adjusted-summand calibrations and the finite prime-strip product formula, while the coric unit-character data constructs the $\Theta^{\times\mu}_{\mathrm{LGP}}$ invariant before entering the existing log-Kummer Ob7 source link. These routes therefore no longer need the determinant/global Ob7 equality as an independent input. This pointwise trace is now itself derived from the existing theta-pilot class-region source once its class formula is identified with the Hodge-theater source spine's theta-pilot class-image family. The route now also accepts record-level theta-pilot class formulas and Hodge-theater/history/log-theta-lattice formulas, projecting them through the class-region source before assembling the Step (xi) comparison. Below the lattice formula layer, it can consume the same-lattice-key image law, which constructs the lattice formula from representative choices before comparison with the source-spine class images. One level lower, it can consume typed theta-pilot fiber-transport invariance and derive the same-key image law internally. Below that, the selected route can now consume the (Ind1), (Ind2) equality-quotient pullback and, on the current strongest local source route, the retained post-procession (Ind2) action-packet transport source; Lean derives the quotient image pullback, the fiber-invariant lattice image law, and the record/source trace before entering the Step (xi) hull/determinant comparison. The action-packet theta-formula comparison with the source-spine class images can now also be derived from the equality of the record possible-image family with the unified Theorem 3.11 source image family; it is no longer a separate input on the exact-theta action-packet route. A further constructed-record variant installs the unified source-spine image family directly as the record's theta-pilot possible-image family, so the record/source image equality becomes definitional; the remaining image-side calibration is that these source-spine images are the package target-region family. This calibration has now been lowered from a whole-family equality to the pointwise theta-pilot class-region trace $\mathcal{R}_{\mathrm{class}}(\mathrm{class}(c)) = \mathcal{R}_{\mathrm{target}}(c)$; Lean derives the source-image/target-region family equality internally before installing the source-spine record. One layer lower, the class-region trace can now be produced from a target-region-backed theta-pilot source: a representative section for theta-pilot classes, a proof that target regions depend only on the theta-pilot class, and explicit target-region point witnesses. Lean constructs the witness-bearing class-indexed possible-image source from these data and threads it through the exact-theta Step (xi) action-packet route. This has now been lowered once more: the target-region class-invariance law can be derived from invariance under the concrete theta-pilot lattice key, namely Hodge theater, history label, log-theta lattice coordinate, and coric datum. This lattice-key boundary has itself now been lowered to a target-region lattice formula: each package target region is identified with a formula indexed by the Hodge theater, history label, theta-pilot lattice coordinate, and coric datum, and Lean derives the key invariance, class trace, full source-image/target-region family equality, and Step (xi) source-spine handoff from that formula. The formula is now also produced from the concrete lattice-image law already present in the Theorem 3.11 source spine: record possible images are evaluated by the lattice formula, and the multiradial record's target-region realization identifies those record regions with the package target regions. The exact-theta Step (xi) source-spine route can therefore consume the lattice-image law directly, without an independently supplied target-region lattice formula. The same package target-region formula is now projected further down from typed theta-pilot fiber transport, from the (Ind1), (Ind2) equality-quotient pullback, and from the retained post-procession (Ind2) action-packet transport source. Thus this target-region formula boundary follows the current deepest local Theorem 3.11 action-packet source rather than stopping at a standalone lattice-image law. The action-packet boundary itself has now been lowered one step: Lean projects the retained packet/action transport source from place-audited nonarchimedean Ism or archimedean order-two action-entry label-transport data, together with explicit identifications of the audited packet tensor states with the post-procession source and target local tensor states. The label-transport source is now constructed one layer lower from the retained action-entry source, source and target tensor symmetry label equations, and transport of the direct-summand symmetry kind. The same tensor-label data now also projects to the place-audited possible-image/fiber audits for the nonarchimedean and archimedean branches, so the lower action data carries both the packet transport and the equality-quotient/fiber witness information. This boundary has now been lowered again: the tensor-label equalities may be read from a single direct-summand packet SymmetryLabelTransportSource, and the corresponding concrete fiber source projects through the tensor-label route internally. Record-level possible-image point witnesses now also factor through the class-indexed Theorem 3.11 possible-image source and the record/source image equality, so the newest symmetry-label fiber source no longer needs independently supplied witness points. The same record/source image equality now gives all-choice nonemptiness for the additive-Haar arithmetic-divisor and record Ob3/Ob5 constructors, and builds the central Theorem 3.11 possible-image source-data record with its nonemptiness audit, rather than requiring a separate downstream nonemptiness field. The exact-theta selected Step (xi) paper-trace constructor now also accepts this symmetry-label transport source directly and projects the older action-packet transport internally; the source-spine record constructor has the same symmetry-label entry point. At the public paper-trace route boundary, the remaining payload audit has now been split so the caller supplies only the non-Step-(xi) payload; the legacy Step-(xi) audit fields are derived internally from the structured Step-(xi) source. A stricter concrete-localized additive-Haar wrapper now makes this split explicit at the normalized Haar-modulus boundary. This has now been tightened again: the public wrapper may take only the seven lower arithmetic/formula fields from Theorem 1.10/IUT IV, and a further wrapper derives those seven fields from the IUT IV C_Θ

source ledger itself. The Step (x) realified-packet field is projected from the vertical log-Kummer package, the IPL/SHE/APT fields are projected from the Theorem 3.11 bridge package, and the Step (xi) arithmetic-degree, localized determinant, adjusted log-volume, and endpoint payload slots are constructed from the normalized finite-summand source-route audit, which explicitly carries the projected additive-Haar, Haar-modulus, and localized-adjusted Hodge-summand expansion boundaries. In the newest normalized additive-Haar route, local additive-Haar normalization, the Theorem 1.10 arithmetic-divisor split, the finite-place Haar defect, the arithmetic-gap estimate, and the local-to-global C_Θ comparison are read from IUTIVCThetaObligations, not supplied as a separate residual record. This CTheta-derived arithmetic boundary has also been lifted through the intrinsic finite-extension, unit-ball Haar-character, and residue-module/unit-ball normalized public routes, so those stronger p-adic endpoints can now use the same source-derived arithmetic payload. The unit-ball normalized wrapper has its own full-boundary audit, and the strongest residue-module/unit-ball wrapper now factors through it before projecting to the intrinsic Haar-modulus normalized route. Its audit combines the Step (xi) residue-module source audit, the transported residue-module factor handoff, the projected unit-ball normalized audit, the IUT IV arithmetic-residual audit, and the definitional equality with the projected normalized public route. The parallel non-normalized unit-ball Haar-character and residue-module/unit-ball Haar-character finite-summand routes now have the same CTheta-derived arithmetic handoff: their public wrappers project to the intrinsic Haar-modulus/additive-Haar finite-summand route, and their full audits record the Step (xi) source boundary, the projected finite-summand boundary, the IUT IV residual audit, and the resulting remaining arithmetic-payload audit. On the hull/determinant side, the Ob3/Ob5 compatibility record can now be constructed from the lower equality $\mu_{\log}(\varphi(P_B)) = \widehat{\deg}_{\mathrm{norm}}$: the concrete quotient theta-pilot constructor builds the canonical hull approximant versus weighted-determinant compatibility internally from that equality. The exact-theta variant also uses $\theta_{\mathrm{signed}} = \mu_{\log}(\varphi(P_B))$ to derive the Ob4 tensor-power bound, so that bound is no longer a separate input on this constructor path. The concrete-localized route wrapper takes the minimal-hull source, localized vector-bundle determinant decomposition, and $\Theta^{\times\mu}$ Ob7 source directly, constructs the paper-derived Step (xi) source internally, and then invokes the preferred route. A stronger hull-side wrapper now goes below the minimal-hull argument: it constructs the concrete hull source internally from the source-spine equality-quotient possible-image family and the principal $\lambda \cdot O$ hull source, then threads that hull source through the localized determinant and Ob7 route. This principal-product route now has a strict variant in which the determinant upper bound is derived from Θ_{signed} equal to the localized family-hull log-volume. The strict principal-product localized wrapper has now been promoted to the named public Step (xi) paper-source boundary. The older route theorems still exist as compatibility layers, but the advertised public audit no longer takes an already constructed Step (xi) source. A product-formula strengthening of this public audit now also constructs the localized $\Theta^{\times\mu}$ Ob7 source from the localized determinant source, local Gaussian summand calibration, finite determinant product formula, and coric unit-character data. Thus the public route can consume product-formula data in place of a prebuilt Ob7 compatibility record. The public audit now has a stricter raw-adjusted variant: it consumes raw adjusted-localization Gaussian calibration data, constructs the localized product-formula source internally, and then invokes the product-formula public route. Each determinant summand is assigned a Gaussian place and raw ratio, the partition weight is computed from the Ob3 localization weight times this ratio, and Lean derives the weighted summand calibration and global product-formula equality from the Gaussian local-global Frobenioid restriction identity. This has now been lowered once more: the public route also accepts a globally normalized raw-adjusted source in which the raw ratio is computed as adjusted raw log-volume divided by the nonzero

Gaussian global log-volume. The Ob3 weighted determinant sum together with determinant/global prime-strip equality proves the finite normalization. The Ob7 prime-strip boundary has also been lowered: common environment and Gaussian prime-place coordinates plus one valuation-indexed coric character now construct the prime evaluation, $\Theta^{\times\mu}$ lift, and coric invariant internally. A further public wrapper constructs the local-global collection, its environment anchor, and the local environment/Gaussian evaluation from global-to-local restrictions and local object identifiers; the local realified objects are forced by the restricted global prime log-volumes. In the diagonal finite valuation-index route, the environment and Gaussian prime-place equivalences are identity maps constructed internally. The preferred non-vacuous public diagonal wrapper now uses the additive-Haar compact-open factor boundary over the principal product hull system, together with an audited Hodge–Arakelov theta evaluation, global-to-local restriction data calibrated at the Hodge canonical full-label Gaussian degree, and the scalar equality between the Hodge theta-monoid degree and the localized adjusted determinant sum. The environment/Gaussian realified Frobenioid evaluation is constructed internally from zero-divisor ordinary/theta/log-shell copies whose unit log-volume is the Hodge canonical degree; normalization proves that the Gaussian theta divisor log-volume is this Hodge canonical degree. The Hodge evaluation identifies the canonical degree with the theta-monoid degree. The finite Hodge summands are no longer supplied as a public family: they are fixed internally to be the localized weighted adjusted determinant contributions, and the scalar equality proves that their finite sum is the theta-monoid degree. The structure-sheaf log-volume is identified with a distinguished direct summand of each localized arithmetic vector bundle. Each direct-summand log-volume is projected from the normalized log-volume of a local compact-open tensor factor constructed from additive Haar data. The additive-Haar layer now has a strict factor constructor: from the source-level inequality $1 < \mu(U)_{\mathbb{R}}$ for the selected non-structure compact open, Lean proves positivity of $\log(\mu(U)_{\mathbb{R}})/[K : \mathbb{Q}_p]$, then uses the norm-square projection to derive the nonzero direct-summand norm. This generic interface has been instantiated by a concrete finite-extension local-field source: the selected factor is the inverse base-prime dilation $p^{-1}O_K$, whose compact-openness and Haar mass $p^{[K:\mathbb{Q}_p]} > 1$ are proved in the source core and threaded through the preferred additive-Haar public wrapper. The same wrapper can now consume a family-hull-indexed calibrated localized hull-cover source, deriving the old family-hull containment, log-volume summation, and pointwise localized-region determinant identities from the index-fiber presentation, finite additivity, and local vector-bundle calibrations. A stricter variant lowers the volume-summation input from a global finite-additive hull law to a cover-local log-volume measure model for the localized regions under construction. The newest direct-product variant also constructs the family-hull index from local factor regions: the direct-product cover identity gives existence of an index, and local factor separation proves the required fiber equality. A tensor-measure refinement constructs the cover-local measure model from direct-product cell log-volumes and the tensor-cell summation formula, so the direct-product public route no longer needs an independently supplied finite-cover measure model. A source-core refinement lowers this again: the tensor cells are now the localized determinant summands from the additive-Haar source-core decomposition, and the tensor-cell finite-sum formula is derived from the source-core localized log-volume theorem together with the direct-product cover identity. The remaining input at this boundary is the geometric identification of the localized regions with the direct-product local-factor cells and their cover of the principal-product family hull. A charted local-ring refinement lowers the factor interface itself: the local factors are no longer an arbitrary region family, but the regions of per-place local-ring/vector-bundle calibration objects whose chart log-volumes are identified with the corresponding direct-summand log-volumes. The cover identity and factor separation are still explicit geometric inputs. A valuation-ball charted refinement now lowers the factor calibration one step further: each local-ring chart is projected from a valuation-ball/additive-Haar factor calibration, and Lean derives the charted local-ring cover interface from the valuation-ball regions, their separation, and the valuation-ball family-hull identity. The p-adic finite-extension branch now has the same principal valuation-ball lowering: its direct-product cell identity, local-factor separation, and family-hull equality are projected from the SourceCore principal valuation-ball product-hull cover before entering the p-adic charted route. The p-adic branch also has principal-hull and hull-formation alignment refinements: the valuation-cover/public-hull equality is derived from the principal valuation-ball hull theorem, and the source-core and valuation-cover possible-region equalities are projected from the concrete `principalProductHullFormationData` family. The p-adic calibration boundary is also lowered: the localized calibration and per-factor valuation-ball calibration are transported from the principal valuation cover across the principal-hull equality, so they are no longer supplied independently at this level. The additive-Haar principal valuation-ball refinement lowers this again: the direct-product cell identity, local-factor separation, and family-hull equality are projected from the SourceCore principal valuation-ball product-hull cover. The selected valuation-ball branch now also transports exact possible-image $\Xi(P_B)$ witnesses across carrier equivalences: Lean proves this first for the holomorphic hull operator and possible-image family, then applies it to the radius-cover exact-Ξ source. A public adapter now aligns this transported source with the Theorem 3.11 quotient family using quotient-index matching, pointwise possible-region equality, and hull-operator equality; the public exact-Ξ log-volume calibration is then derived rather than supplied. The quotient-index matching has also been lowered to concrete choice-level data: a selected valuation-ball index attached to each Hodge-theater/log-theta choice, invariance under the Theorem 3.11 equality quotient, and representatives over selected indices construct the quotient equivalence used by the adapter. Thus the remaining boundary at this additive level is to feed this choice-level adapter from the strongest preferred route while continuing to lower the additive-Haar source-core localized determinant decomposition. A principal-hull-aligned refinement removes the direct valuation-cover/public-hull equality from this boundary: Lean derives it from the lower principal valuation-ball cover's own principal-hull theorem and the equality between that principal hull system and the public Theorem 3.11 principal-product hull system. A hull-formation possible-region refinement lowers the remaining possible-region equality: the source-core and valuation-cover families are aligned with the concrete family inside 'principalProductHullFormationData', and Lean derives the public 'principalProductPossibleRegion' equality by projection. The p-adic source-core localized determinant alignment is now constructed from these transported valuation-cover fibers: the finite-extension source core is built from transported local calibration data and finite-extension unit-ball Haar factors, and Lean proves the projected vector-bundle and localized-region equalities consumed by the older route. The remaining lower matching datum is the factor-level equality between each projected finite-extension valuation ball and the transported valuation-cover factor. A finite-extension-backed valuation-cover wrapper now lowers the valuation-cover side itself: the lower principal valuation cover is projected from finite-extension p-adic Haar data before it enters the constructed source-core route. A hull-formation-selected intrinsic cover constructor fixes the lower p-adic cover's hull system to the public principal-product hull system and fixes its possible-region family to `principalProductHullFormationData`; the intrinsic possible-region comparison is therefore produced by construction, while the principal-hull comparison is exposed as the lower equality between the p-adic principal hull source and the public principal-product hull system. A public-principal specialization now constructs this lower p-adic principal source as the `ULift` copy of the public principal-product source and proves the product-hull intersection law by transport through the public intersection theorem. Lean now also proves that the induced holomorphic hull system is unchanged by this `ULift`: the proof reduces equality of `IUTStage1Remark395HolomorphicHullSystem` records to equality of the `isHull` predicate and `logVolume` field via the generated constructor-injectivity theorem. The intrinsic wrapper also removes the separate public finite-extension factor family: Lean transports the lower cover's finite-extension factors across the same hull-system equality used for the valuation-cover calibration fiber and derives the old factor-matching field internally. An inverse-dilation refinement lowers the analytic input further: each transported valuation-ball Haar factor is identified with the inverse base-prime unit ball of a constructed finite-extension p-adic Haar source, and Lean derives $1 \leq \mu(U)_{\mathbb{R}}$ from the strict mass theorem $1 < \mu(p^{-1}O_K)_{\mathbb{R}}$. The strict selected determinant summand is now tied to this same inverse-dilation family: the unified public route projects the selected summand from the (`positiveIndex`, `positiveSummand`) member of the valuation-cover family, rather than accepting an independent selected-factor inverse-dilation source and equality. A boundary audit theorem now exposes this dependency shape by proving both the all-factor and selected-summand inverse-base-prime unit-ball equalities from the same family. The hardened preferred wrapper lowers the theta scalar input one more step: it takes the Hodge/family-hull equality and derives the localized adjusted determinant-sum equality by the localized hull decomposition theorem before invoking the unified inverse-dilation route. This wrapper has now been lowered again: the inverse-dilation branch can start from the finite localized Hodge-summand formula and derive the Hodge/family-hull comparison from the same Remark 3.9.5 decomposition. Lean also proves the converse source-facing expansion needed by the public route: a localized-adjusted inverse-dilation source projects to the finite Hodge-summand source by unfolding the localized adjusted sum as the finite sum of weighted adjusted log-volumes. This expansion is now also available from the constructed Gaussian/Hodge inverse-dilation wrapper itself: Lean projects its localized-adjusted theta equality to the lower source and then invokes the finite summand route. The Haar-modulus-backed inverse-dilation wrapper now exposes the same finite-summand expansion boundary after projecting its measure-level inverse-base-prime cover to the generic inverse-dilation source. The additive-Haar construction layer now uses this same route after it constructs the Gaussian/Hodge localized-adjusted source from compact-open factors and the strict inverse-base-prime mass theorem. The normalized structure-sheaf branch now uses the finite-summand route as well: after deriving the direct-summand handoff from compact-open normalized Haar data, the residual and IUT-IV public wrappers enter through the Hodge-summand expansion endpoint rather than the older family-hull endpoint, and the additive-Haar arithmetic-degree payload is certified by the same normalized finite-summand audit. The non-normalized additive-Haar/Gaussian-Hodge companion now has the parallel summand audit and an IUT-IV arithmetic-source wrapper for the finite-summand endpoint, so its determinant/calibration payload is no longer tied only to the older family-hull route. The same finite-summand endpoint has now been lifted through the intrinsic finite-extension Haar-modulus, unit-ball Haar-character, and residue-module quotient wrappers, so those local arithmetic sources no longer expose only the family-hull route at their public boundary. On the normalized branch, the unit-ball Haar-character wrapper now has its own CTheta-derived endpoint and full boundary audit, and the residue-module normalized route now factors through that unit-ball audit before the intrinsic Haar-modulus normalized boundary. The non-normalized unit-ball Haar-character wrapper and the residue-module/unit-ball Haar-character wrapper now also have CTheta-derived finite-summand endpoints and full boundary audits, so their remaining additive-Haar arithmetic payload is projected from the IUT IV ledger rather than supplied separately. It now also lowers the structure-sheaf input: the hardened wrapper takes localized direct-summand calibration and derives the additive-Haar normalized compact-open equality by the additive-Haar projection theorem. A complementary non-vacuous Haar-modulus wrapper now uses the opposite, source-level direction required for the additive-Haar construction: it takes the localized structure-sheaf equality with the normalized compact-open Haar factor and derives the Step (xi) structure-sheaf/direct-summand equality from the additive-Haar localized vector-bundle decomposition theorem. The current preferred wrapper lowers the surrounding Gaussian/Hodge boundary again: the restriction local-global object, ordinary-log-volume/Hodge-degree calibration, direct-summand data, and Hodge/family-hull equality are projected from the constructed Gaussian/Hodge localized-adjusted source. It now lowers that source boundary further: Lean constructs the Gaussian/Hodge localized Ob7 source from additive-Haar compact-open factors, derives the selected strict mass from the inverse base-prime factor, projects the Haar-modulus valuation-cover side from the intrinsic finite-extension valuation-cover source, and derives the inverse-dilation Haar-modulus family from unit-ball Haar-character data. The latter is now itself constructed from residue-module quotient data: O_v/p_vO_v, $\#\kappa_v = p_v$, and $\dim_{\kappa_v}(O_v/p_vO_v) = [K_v : \mathbb{Q}_p]$ give the coset decomposition and $\mu(p_vO_v) = p_v^{-[K_v:\mathbb{Q}_p]}$. The normalized-Haar direct-summand handoff has now been propagated through this intrinsic finite-extension and residue-module route as well: the residue source supplies the structure-sheaf equality with the normalized additive-Haar compact-open factor, and Lean derives the Step (xi) structure-sheaf/direct-summand equality before invoking the non-vacuous comparison route. The transported-factor handoff audit follows the residue-module source into the localized additive-Haar vector-bundle factor: the transported valuation-ball factor and the localized determinant factor are both identified with the inverse base-prime unit ball $p^{-1}O_K$ constructed from the residue quotient data. Lean now also projects this equality to the concrete scale invariants used downstream: Haar mass $p^{[K:\mathbb{Q}_p]}$ and strictly positive normalized log-volume for both factors. This handoff is threaded through the additive-Haar residue audit, the normalized residue audit, and the strongest CTheta full-boundary audit, so the public audit object exposes the determinant-factor match directly. The equality boundary has also been lowered in two stages. A component-matched residue source asks for equality of the defining additive-Haar compact-open data fields, and Lean reconstructs the previous full source equality by additive-Haar extensionality. Below that, a scale-component residue source matches the transported factor directly against the finite-extension inverse-dilation mass source: local ring, residue prime, finite degree, realization, Haar measure, $p^{-1}O_K$, and base-prime dilation. A further valuation-ball source now lowers this once more: the transported local factor is matched before additive-Haar projection, including its valuation norm, selected valuation ball $\{x \mid v(x) \leq r\} = p^{-1}O_K$, Haar measure, local realization, and pointwise base-prime dilation. Lean derives the previous scale-component source from this valuation-ball match, and then projects through the unit-ball Haar-character and intrinsic Haar-modulus wrappers, so the lower valuation ladder consumes the same concrete inverse-dilation components rather than a fresh equality at each layer. The valuation-ball source itself has also been lowered into SourceCore: from the constructed finite-extension dilation-mass source and a positive-radius identification $p^{-1}O_K = \{x \mid v(x) \leq r\}$, Lean constructs the valuation-ball additive-Haar source and proves the inverse base-prime component match. Thus this part of the interface no longer needs to be entered as a single opaque equality of normalization records. Under the normed-algebra compatibility for a finite extension $K/\mathbb{Q}_p$, the radius identity itself is now derived: Lean proves from $\|\mathrm{algebraMap}(p)\| = \|p\|_p = p^{-1}$ that the inverse base-prime unit ball is exactly the valuation ball of radius p, and constructs the corresponding inverse base-prime valuation-ball source. This radius computation also exposed a hard obstruction in the surrounding Step (xi) interface: if the transported intrinsic finite-extension factor is still the valuation-unit-ball cover, then its selected radius is 1, while the constructed inverse base-prime factor has radius p. Lean proves that the corresponding normed finite-extension match source is locally contradictory, via `NormedFiniteExtensionInverseBasePrimeValuationBallMatchedSource.no_unitBallCover_factor`. The replacement local cover has now been added: SourceCore constructs a finite-extension inverse-base-prime valuation-cover packet whose local selected factors have radius p from the outset, projects it to the generic valuation-ball tensor packet, and proves the inverse-base-prime component match for every local factor. Step (xi) also has a hull-formation-selected public principal-product sibling route for this radius-p cover, so the unit-ball branch and the inverse-dilated branch are separated explicitly. The residue-module handoff has now been moved onto this inverse cover at the local-factor level: the new handoff identifies the cover's inverse-dilation mass source with the residue-module constructed mass source and derives the equality between the radius-p valuation factor's additive-Haar normalization and the residue-module inverse-base-prime unit-ball normalization. The preferred-public normalized additive-Haar route has now been migrated to this inverse-cover projection as well: its Haar-modulus valuation-cover argument is definitionally the radius-p residue-module inverse-cover source, rather than the legacy unit-ball intrinsic cover. The non-normalized additive-Haar/Gaussian-Hodge route has now been added for the same inverse-cover source: the residue-module wrapper supplies the localized structure-sheaf/direct-summand comparison, projects the Haar-modulus valuation-cover argument from the radius-p inverse cover, and then reuses the generic additive-Haar compact-open ladder to construct the Gaussian/Hodge localized-adjusted Ob7 source. Thus the normalized public wrapper is no longer the only residue-module inverse-cover entry point. The additive-Haar arithmetic-degree obligation package for this non-normalized route is now constructed as well: its Step (xi) determinant/calibration slots are read from the additive-Haar/Gaussian-Hodge finite-summand source audit, and the remaining lower arithmetic residual is projected from the IUT IV C_Θ ledger. This boundary has been lowered once more: the residue inverse-cover route can now take a constructed Gaussian/Hodge localized-adjusted source over the same residue-derived localized hull vector bundle, and Lean projects from it the Hodge local-global comparison, selected direct-summand data, structure-sheaf equality, and localized-adjusted theta equality required by the older additive-Haar route. The same residue corridor now has a summand-form source: the localized-adjusted theta equality is derived by summing the finite Hodge summands and identifying them pointwise with localized weighted-adjusted determinant terms. A companion restriction-local-global wrapper now derives the older public source's global object, localization, local object identifiers, and pointwise Hodge/local-prime equality from the environment-anchored realified Frobenioid restriction source. The constructed Gaussian/Hodge residue source now also factors through this restriction wrapper, so its anchored local-global source is produced by the constructed summand source rather than bypassed by the older public fields. What remains below this route is to construct the finite Hodge-summand formula itself from the lower Hodge-theater and log-theta-lattice data. The full-boundary audit for the current route also ties the IUT IV arithmetic residual to the same public wrapper. The remaining principal-hull boundary is now to derive the coordinate scalar-image landing/preimage laws for the generic public principal-product hull source from Hodge-theater/log-theta-lattice data. For the selected valuation-ball hull, the nonzero-scalar multiplication route now constructs the selected scalar-parameter source and its Ob3/Ob5 handoff; the remaining work is to construct the lower transported-factor/residue-module matching data directly from the same Hodge-theater/log-theta local-field package. The valuation-unit-ball finite-extension branch remains only a normalized legacy boundary: the selected radius is forced to be 1, so $\mu(O_v) = 1$ constructs weak nonnegativity but also proves that the normalized log-volume of the selected unit-ball factor is 0. Thus the strict-positive unit-ball route is explicitly quarantined as vacuous, now also at the exact intrinsic finite-extension public endpoint. Finite summation derives the Ob3-1 inequalities comparing the structure-sheaf log-volume with the localized bundle log-volume; the subtraction formula derives adjusted raw local positivity; positive determinant weights derive localized adjusted summand positivity; finite summation gives localized family-hull positivity. The theta scalar at the strict boundary is the environment theta divisor log-volume by definition. The public route now constructs the older Hodge-summand, Gaussian/Hodge, Hodge-evaluation, and Hodge-scalar sources internally from the Hodge canonical-degree divisor construction, audited evaluation, and localized adjusted-sum calibration, then derives the environment-theta/localized adjusted-sum equality; Lean then derives equality with the family-hull log-volume from the localized Ob3/Ob5 identity. This derived equality implies the family-hull/global calibration via theta/ordinary realified Frobenioid compatibility, and the localized determinant/family-hull equality then gives the determinant/global equality needed for the Gaussian-global nonzero proof. The earlier structure-sheaf/direct-summand boundary has also been lowered at the finite-extension endpoint: instead of publicly supplying 'structure-sheaf equals direct summand', the newest compact-open factor route uses the valuation-unit-ball normalized log-volume identity for the distinguished structure-sheaf factor and derives the direct-summand equality. The non-vacuous Haar-modulus/additive-Haar branch now has the same normalized-Haar-to-direct-summand handoff without passing through the vacuous unit-ball endpoint. Its preferred boundary audit also exposes the Step (xi-a)–(xi-g)/Ob1–Ob9 paper trace constructed from the same additive-Haar compact-open factor source, including the Ob7–Ob9 log-Kummer/vertical-shift/realified-log-volume entries. For strict positivity, the additive-Haar route now has a genuine Haar-mass-to-norm constructor, and the inverse-dilation finite-extension branch constructs a non-unit selected factor with strict Haar mass. The unit-ball finite-extension route remains as an explicit vacuity diagnostic. The earlier structure-sheaf-positive, raw-positive, summand-positive, explicit-ratio, weighted-partition, and one-summand localized constructors remain as compatibility layers. This is progress below a free-floating obligation bundle. What remains is to construct, from the source papers, the actual lower operations corresponding to:

the analytic construction of the class-indexed possible-image source,
the analytic construction of the place-audited packet/action symmetry-label transport inputs,
the analytic construction of the global-to-local restrictions and local-prime global equality,
the analytic construction of non-diagonal prime-place coordinates and the coric character,
the Hodge canonical-degree local-prime and localized adjusted-sum calibration,
the construction of the remaining theta-region local-field Haar factors,
and their calibration from the Hodge-theater packet data.

The strict route now derives the unit-ball tensor factors and one non-unit inverse-dilation strict factor from finite-extension local-field Haar data, and can build the record Ob3/Ob5 determinant/log-volume source from a principal-product theta-region cover identity and its finite additivity formula. The remaining work is to construct the full analytic valuation-cover family, its global bridge calibrations, and the Hodge-theater packet source that selects the appropriate non-unit local factors.

5.3 IUT IV estimates

The C_Θ comparison depends on IUT IV local and global estimates. The current formalization organizes the route through additive-Haar arithmetic-degree and p-adic formula objects, but the deepest estimates are still source obligations. What is now checked for the preferred route is the ordered-real and finite-sum handoff from the arithmetic gap to the canonical C_Θ-scale comparison, together with the source-level bridge that identifies the constructed Step (xi) canonical scale with the finite-place IUT IV scale before applying plus-one cancellation. What remains to construct from first principles is the lower IUT IV source of that arithmetic gap:

the distinguished log-shell inclusions and numerical bounds,
the nondistinguished zero log-volume contribution,
the archimedean metric containment,
the arithmetic divisor source for Theorem 1.10,
the distinguished and archimedean formula-to-gap bounds,
the finite-place summation comparison.

Until these estimates and the upstream source spine are constructed from the published source data, the corridor remains conditional on source-paper mathematical content.

5.4 Local arithmetic and p-adic analytic infrastructure

The implementation has moved below a purely formal real-number boundary, but it still lacks significant local arithmetic infrastructure. The remaining work includes finite extensions of $\mathbb{Q}_p$, residue fields, Haar-normalized unit balls, explicit local degrees, different and conductor terms, and the formula matching that connects those objects to the arithmetic-degree route. This is standard mathematical infrastructure in spirit, but it is not small work in Lean.

6 Blueprint and Apodeixis

6.1 Lean blueprint status

The repository contains a Lean blueprint under `blueprint/`. It has a compact prose structure: aim and scope, the source Stage 1 route, the formal interface, the route to Corollary 3.12, open source obligations, and later work. It also has a generated declaration index, `blueprint/lean_decls`, which currently lists thousands of Lean declarations.

This means the blueprint is not empty, but it is still intentionally slim. It is adequate as a navigational document for the present stage. It is not yet a complete dependency graph of the project in the strongest sense: many nodes are marked not ready, and the prose chapters do not yet expand every major Lean route into a polished theorem-by-theorem mathematical dependency tree.

The right standard for the blueprint is different from the standard for this paper. The blueprint should be a map: concise nodes, Lean links, readiness status, and dependencies. This paper should be an argument: what we are formalizing, why the type boundaries matter, what is proved, and what remains mathematically open.

6.2 Apodeixis integration plan

The project is intended to be compatible with Apodeixis as a collaboration and review layer. The practical goal is to expose the formalization in a way that supports external mathematical review:

$$\begin{gathered}\text{source-paper claim}\\ \Downarrow\\ \text{blueprint node}\\ \Downarrow\\ \text{Lean declaration or source obligation}\\ \Downarrow\\ \text{audit status and review discussion.}\end{gathered}$$

For this to work, the Lean code must avoid opaque omnibus assumptions, the blueprint must keep source claims linked to declarations, and the paper must avoid becoming a changelog. The current audit records are designed with this future use in mind. They give Apodeixis a natural unit of review: a named mathematical obligation with a precise Lean consumer.

7 Methodological Lessons

7.1 Typed boundaries are productive

The main technical lesson so far is that typed boundaries are not bureaucracy. They are mathematical tests. Once ordered real-line copies, transports, finite-label quotients, hull regions, source certificates, and upper-semi relations are assigned different types, informal shortcuts become impossible unless they are backed by explicit constructors.

This is especially useful in the 3.11–3.12 corridor because the public dispute is about whether a comparison has lost information during transport between different mathematical histories. Lean cannot decide that philosophical issue by itself, but it can prevent a formal proof from hiding the relevant movement.

7.2 Conditional routes are still useful

A conditional theorem can be scientifically useful if it names its conditions precisely. The current Stage 1 route is not a proof of Corollary 3.12 from the published IUT papers, but it has lowered the ambiguity of the problem. The remaining obligations are no longer "somewhere in the comparison". They are organized into Theorem 3.11 source construction, Step (xi) hull/determinant construction, IUT IV local-to-global estimates, and local arithmetic infrastructure.

7.3 The paper must not be a ledger

The earlier draft recorded too many implementation increments. That made it long but not more informative. The standing policy for future updates should be:

update the relevant section,
replace stale claims,
move only stable mathematical outcomes into the paper,
leave declaration-level detail to Lean and the blueprint.

This paper should remain regular research-paper length. The blueprint and generated declaration index are the correct place for detailed navigation.

8 Roadmap

The next milestones should eliminate source-obligation families rather than adding more endpoint wrappers.

Milestone A: Step (xi) hull/determinant construction. Replace the current Remark 3.9.5/Step (xi) obligation fields by constructed holomorphic hull and determinant data. This is the most direct way to advance the 3.11–3.12 issue, because it targets the passage from multiradial possible images to the signed q-region comparison.

Milestone B: IUT IV local-to-global C_Θ. Construct the local estimates and finite-place summation chain underlying the canonical C_Θ-scale comparison. This would remove the deepest remaining numerical assumption in the current endpoint.

Milestone C: Theorem 3.11 source derivation. Work backward from the typed indeterminacy core to actual Hodge-theater, Frobenioid, log-Kummer, and log-theta-lattice source data. This is the largest milestone and the one most directly connected to the Scholze–Stix objection.

Milestone D: Blueprint expansion. Expand the blueprint from a slim guide plus declaration index into a richer review graph. Each nontrivial source obligation should have a node with source citations, Lean declarations, readiness status, and a short explanation of the mathematical construction still missing.

9 Conclusion

The project has reached a useful but conditional Stage 1 state. Lean verifies the final real inequality, distinguishes the disputed comparison levels, keeps $\mathrm{Ind}_1/\mathrm{Ind}_2$ equality behavior separate from Ind_3 upper-semi behavior, and threads a concrete Hodge-theater/log-theta-style source route into the current C_Θ/Corollary 3.12 endpoint. The code is not merely a prose annotation of the papers; it is an executable audit of a precise corridor.

What remains is real mathematics, not formatting. To settle the 3.11–3.12 issue inside this project, the source obligations must be discharged from Mochizuki's constructions: full Theorem 3.11 multiradial data, genuine Remark 3.9.5 hull/determinant operations, IUT IV local estimates, and the local arithmetic infrastructure connecting them. The formalization has made that task sharper. It has not made it disappear.

References

[1] S. Mochizuki, *Inter-universal Teichmuller Theory I: Construction of Hodge Theaters.*

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[3] S. Mochizuki, *Inter-universal Teichmuller Theory III: Canonical Splittings of the Log-theta-lattice.*

[4] S. Mochizuki, *Inter-universal Teichmuller Theory IV: Log-volume Computations and Set-theoretic Foundations.*

[5] S. Mochizuki, *On the Formalization of IUT: Preliminary Progress Report*, 2026.

[6] P. Scholze and J. Stix, *Why abc is still a conjecture.*